Geometry : Vectors — problem 12.1.22

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Consider n vectors $\vec{a}_1, \ldots, \vec{a}_n$, $|\vec{a}_i| \leq 1$. Show that, in the sum $\vec{c} = \pm\vec{a}_1 \pm \cdots \pm \vec{a}_n$, one can choose the signs so that $|\vec{c}| \leq \sqrt{2}$.